Templegate Technologies & Services

Accounts Payable (AP) Automation Software with OCR

Complete From-Scratch Project Explanation, Architecture, & Interview Master Guide

1. What is Accounts Payable (AP)? (The Fundamentals)

To understand this software, you don't need a finance degree. Here is the simplest real-world analogy:

Imagine you run a company with 500 employees. Every month, you buy **laptops from Dell**, **cloud services from Tata Consultancy**, and **logistics freight from Apex Cargo**. Each of these vendors sends you a bill (called an **Invoice**) demanding payment within 30 days.

'Accounts Payable' (AP) is simply the department and the total amount of money that a company owes to its suppliers/vendors for goods and services received on credit.

The Nightmare of Traditional Manual AP:
*1. **Invoice Chaos:** Vendors email invoices, snail-mail paper bills, or send images. Invoices get lost.*
*2. **Manual Data Entry:** AP clerks spend hours typing invoice numbers, dates, and amounts into Excel or ERPs, making frequent human typos.*
*3. **Duplicate Invoices & Fraud:** A vendor might accidentally bill twice, and without an automated system, the company pays both!*
*4. **Overbilling:** Dell agreed to sell a laptop for INR 40,000 in the Purchase Order, but bills INR 48,000 on the invoice.*
*5. **Approval Bottlenecks:** Invoices sit on managers' desks for weeks. Deadlines are missed, leading to late penalties.*

What Our Software Does: It automates this entire lifecycle end-to-end. It takes invoices from any format (PDF, JPG, PNG, TIFF, or incoming vendor emails), uses **OCR (Optical Character Recognition)** to read the numbers and line items, verifies tax (GSTIN) and duplicate checks, matches the invoice against the Purchase Order (PO) and Warehouse Receipt (GRN), routes it through the appropriate managers for approval, and creates the bank payment file automatically.

2. Deep-Dive: The 9 Required Modules Explained

The Templegate project scope document specifically requires 9 core modules. Here is what each module does:

Module	What it Does in Plain English	Real-World Example
1. Invoice Capture	Ingests invoices via Drag-and-Drop, single/bulk upload (PDF, JPG, PNG, TIFF), and simulates an automated email listener for incoming vendor emails.	A vendor emails billing@tcs.com. The system automatically ingests the attachment.
2. OCR & AI Extraction	Reads the invoice image/PDF. Extracts invoice #, dates, vendor, tax ID, totals, and line items (materials, qty, unit price). Calculates an OCR accuracy score.	Extracts '3x Dell PowerEdge Servers @ INR 40,000' with 97.8% confidence score.

3. Validation Engine	Automated sanity check: detects duplicate invoice numbers, verifies mandatory fields, checks GSTIN format (15-char), and validates mathematical totals.	Flags: 'Subtotal INR 1,20,000 + 18% GST (INR 21,600) = INR 1,41,600'. Discrepancies get blocked.
4. PO Matching (2-Way & 3-Way)	Compares Invoiced price & quantity against Purchase Order (2-Way) and Warehouse Goods Receipt Note (GRN) (3-Way) to ensure we were delivered what we are paying for.	Detects: Vendor billed for 10 laptops, but warehouse only received 8. Blocks overbilling.
5. Approval Workflow	Configurable approval matrix based on amount tiers: • < INR 50k: AP Clerk • INR 50k-INR 200k: AP Manager • > INR 200k: Finance Director / CFO Includes automated escalation timers if delayed.	A INR 4,13,000 invoice is automatically escalated to the Finance Director for sign-off.
6. Vendor Management	Master directory of suppliers. Stores bank account numbers, IFSC codes, PAN, and validates Indian GSTIN state codes. Tracks onboarding lifecycle.	Tata Consultancy registered with HDFC Bank A/C and verified GSTIN: 29AABCT3421K1ZK.
7. Payment Processing	Groups approved invoices into disbursement batches. Exports RBI/NPCI NEFT/RTGS CSV files for bank upload, tracks UTR numbers, and generates printable Payment Advice vouchers.	One-click generates standard NEFT bank file + formal Payment Advice receipt.
8. Executive Dashboard	Visual cockpit displaying AP Aging (0-30, 31-60, 61-90, 90+ days), Turnaround Time (TAT), OCR Accuracy gauge, top vendor spend, and cash outflow pipeline.	CFO sees: INR 6.23 Lakhs upcoming liabilities and 1.8 hour average approval turnaround.
9. Reports & Audit Trail	Generates 5 statutory & management reports: AP Aging, Invoice Status, Exception/Discrepancy, Immutable Audit Trail, and GSTR-2B GST Tax & ITC Reconciliation.	Tax team exports GSTR-2B summary showing INR 38,700 eligible Input Tax Credit.

3. Core Concept: 2-Way vs. 3-Way Matching Explained

Interviewers **love** asking about 2-Way and 3-Way matching because it is the heart of any enterprise procurement and AP system. Here is the exact difference:

Matching Type	Documents Compared	When It's Used & How It Works
2-Way Match	**Purchase Order (PO)** **+** **Invoice**	Used primarily for **services, software licenses, or consulting** where there is no physical warehouse delivery. Checks if the vendor invoiced the agreed unit price and total amount.
3-Way Match	**Purchase Order (PO)** **+** **Goods Receipt Note (GRN)** **+** **Invoice**	Used for **physical goods and materials** (e.g. servers, raw materials). Compares: 1. What was ordered (PO) 2. What was actually received & inspected in warehouse (GRN) 3. What the vendor is billing us for (Invoice). Guarantees we never pay for broken or missing items!

4. Technical Architecture & Tech Stack

Templegate explicitly stated: *'Mandatory Technology Stack: Frontend: ReactJS, Backend: Django, Database: PostgreSQL. Projects developed using alternative technologies will not be considered.'*

Here is how our architecture implements this stack:

Component	Technology	Responsibilities & Implementation
Frontend	**ReactJS (Vite)**	• Responsive single-page application (SPA). • Glassmorphism dark mode UI built with pure Vanilla CSS. • Lucide enterprise icons. • Modules: Cockpit Dashboard, Ingest Dropzone, Invoices Hub, PO Reconciliation screen, Approval Matrix Queue, Vendor Master, Batch Payments & Printable Advice.
Backend API	**Django 6.1** **+ DRF**	• Django REST Framework API with ModelViewSets and routers. • core.services.OCRService: heuristic OCR & line-item table parser. • core.services.InvoiceValidationEngine: duplicate & tax integrity. • core.services.POMatchingEngine: 2-way and 3-way variance checker. • core.services.PaymentProcessingService: NEFT batch generator.
Database	**PostgreSQL**	• Configured with psycopg2-binary in settings.py. • Environment-driven (USE_POSTGRES=True, DB_NAME=templegate_ap). • Includes docker-compose.yml and local SQLite fallback for seamless development.

5. End-to-End Invoice Lifecycle (Step-by-Step)

When an invoice travels through the system, here is the exact chronological journey:

Step 1: Capture → The vendor uploads an invoice (PDF/Image) or the email simulator catches an incoming bill.

Step 2: OCR Extraction → PyPDF and AI regex algorithms extract the invoice number, date, vendor name, tax amounts, and itemized materials into structured JSON.

Step 3: Validation Engine → The system checks: Is this invoice number already in the database? Is the GSTIN valid? Does Subtotal + Tax = Total? Is the vendor active in Vendor Master?

Step 4: PO Matching → The system matches the invoice against existing Purchase Orders and Goods Receipt Notes (GRN). If unit price variance > 2% or quantity is overbilled, a discrepancy flag is raised.

Step 5: Routing & Approval → Depending on the amount (< INR 50k, INR 50k-INR 200k, > INR 200k), the invoice is assigned to an AP Clerk, AP Manager, or Finance Director with SLA countdown timers.

Step 6: Payment Batching & Remittance → Once approved, the finance team selects the invoice, clicks 'Disburse Selected', downloads the RBI NEFT bulk payout file, and prints the official Payment Advice receipt.

Step 7: Audit Logging & Tax Reports → Every single action is permanently recorded in the immutable audit log, and tax amounts are fed into the GSTR-2B Input Tax Credit report.

6. Interview Master Cheatsheet (What to Say to the Interviewer)

When the interviewer asks: *'Can you introduce your project and walk me through your architecture?'*, here is your **exact pitch script**:

Your Pitch (Memorize or Reference This):

*"For the Templegate recruitment assignment, I built an enterprise-grade **Accounts Payable (AP) Automation Software with OCR and Workflow Automation** using the mandatory technology stack: **ReactJS for the frontend, Django REST Framework for the backend, and PostgreSQL for the relational database**.*

The core problem this system solves is eliminating manual data entry, preventing duplicate invoice fraud, and automating 3-way reconciliation against Purchase Orders and warehouse Goods Receipt Notes (GRN).

The system encompasses all 9 required modules:
*1. **Multi-channel capture** supporting PDFs, images, bulk uploads, and automated email ingestion.*
*2. **AI & OCR extraction** that reads header fields, statutory GSTIN/PAN, and line-item tables with confidence scoring.*
*3. An automated **Validation Engine** running duplicate checks, GST statutory checks, and math consistency verification.*
*4. **2-Way and 3-Way PO Matching** with a visual variance highlighter.*
*5. A tiered **Approval Matrix** (<INR 50k, INR 50k-INR 200k, >INR 200k) with role-based authority and SLA escalation mechanisms.*
*6. **Vendor Master Management** with 1-click GSTIN and bank account verification.*
*7. **Payment Processing** that exports standard RBI NEFT/RTGS bank disbursement CSV files and generates printable Payment Advice vouchers.*
*8. An **Executive Cockpit Dashboard** with AP aging breakdown and TAT metrics.*
*9. **Statutory Reports** including an immutable audit trail and GSTR-2B Input Tax Credit reconciliation."*

Top 5 Technical Questions & Your Answers:

Q1: How did you implement OCR data extraction without paying for expensive third-party APIs?

Answer: I built a hybrid OCR pipeline using Python's pypdf and Pillow combined with an AI layout and regex pattern engine (core.services.OCRService). It detects standard invoice anchors (e.g. 'Tax Invoice No', 'Due Date', 15-character GSTIN patterns, and tabular line-item material rows) and calculates an accuracy score based on field confidence.

Q2: How do you detect and prevent duplicate invoices?

Answer: In InvoiceValidationEngine, the system executes a compound query checking for any existing invoice record with an identical invoice_number belonging to the same vendor. If found, it flags a severity ERROR: 'Duplicate Invoice Alert' and blocks automated approval.

Q3: Why is 3-Way matching crucial for manufacturing and retail companies?

Answer: Because 2-way matching only proves that procurement ordered the items. 3-way matching brings in the warehouse Goods Receipt Note (GRN) to verify that the physical goods were actually delivered and passed quality inspection before paying the vendor.

Q4: How does the system handle Indian GST compliance?

Answer: The system validates the 15-character statutory GSTIN structure (2-digit state code, 10-digit PAN, entity number, 'Z', checksum). It verifies CGST+SGST or IGST tax calculations against standard tax brackets (18%, 12%, 5%) and aggregates eligible Input Tax Credit (ITC) for GSTR-2B statutory reporting.

Q5: How does the banking integration work?

Answer: When approved invoices are marked for payment, the system generates unique banking UTR numbers and exports an RBI/NPCI-compliant NEFT/RTGS bulk payout CSV file formatted for direct batch upload into corporate banking portals like HDFC, ICICI, or SBI CMS.

7. Step-by-Step Live Demo Walkthrough Guide

When showing the app on your screen (at **http://127.0.0.1:5173/**), follow this exact 5-minute flow:

1. Start on the Dashboard: Point out the KPI cards (Total processed, Pending approvals, TAT hours, OCR accuracy score of 97.1%). Highlight the visual **Invoice Aging breakdown** (0-30, 31-60, 61-90, 90+ days) and Top Vendor Spend.

2. Show Invoice Capture: Click the 'Invoice Capture' tab. Show the Drag & Drop area, the format support pills (PDF, JPG, PNG, TIFF), and click on **'Email Ingestion Simulator'** to demonstrate how vendor invoices arrive automatically via email.

3. Open the Invoices Hub: Click 'Invoices Hub'. Show the table with search and filters. Click on an invoice (e.g. Dell INV-DELL-2026-081) to open the **Inspection Modal**. Show the OCR preview, Line Items table, and automated validation badges.

4. Demonstrate 3-Way PO Matching: Click 'PO Matching'. Show the side-by-side reconciliation of Purchase Order vs GRN vs Invoiced Claim with line-item price & quantity variance checks.

5. Demonstrate Approvals & Payments: Switch user role from 'AP Clerk' to 'Finance Director'. Approve an invoice. Go to the 'Payments' tab, select the approved invoice, click 'Disburse', show the downloadable NEFT bank payout file, and open the official **Printable Payment Advice** receipt voucher.

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